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Course: OS Lab
Prn : 123B1B241

NON Preemptive:

```
#include <iostream>
using namespace std;

struct Process {
    int pid;
    int arrival;
    int burst;
    int priority;
    int completion;
    int turnaround;
    int waiting;
    bool completed;
};

void swap(Process &a, Process &b) {
    Process temp = a;
    a = b;
    b = temp;
}

void sortByArrival(Process p[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (p[j].arrival > p[j + 1].arrival) {
                swap(p[j], p[j + 1]);
            }
        }
    }
}

void inputProcesses(Process p[], int n) {
    for (int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        cout << "\nProcess " << p[i].pid << endl;
        cout << "Arrival Time: ";
        cin >> p[i].arrival;
        cout << "Burst Time: ";
        cin >> p[i].burst;
        cout << "Priority: ";
        cin >> p[i].priority;
    }
}
```

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```
        p[i].completed = false;
    }
}

void priorityScheduling(Process p[], int n) {

    sortByArrival(p, n);

    int currentTime = 0;
    int completedCount = 0;

    while (completedCount < n) {

        int idx = -1;
        int highestPriority = 9999;

        for (int i = 0; i < n; i++) {
            if (!p[i].completed && p[i].arrival <= currentTime) {

                if (p[i].priority < highestPriority) {
                    highestPriority = p[i].priority;
                    idx = i;
                }
                else if (p[i].priority == highestPriority) {
                    if (p[i].burst < p[idx].burst) {
                        idx = i;
                    }
                }
            }
        }

        if (idx != -1) {
            currentTime += p[idx].burst;
            p[idx].completion = currentTime;
            p[idx].turnaround = p[idx].completion - p[idx].arrival;
            p[idx].waiting = p[idx].turnaround - p[idx].burst;
            p[idx].completed = true;
            completedCount++;
        } else {
            currentTime++;
        }
    }
}
```

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```
void displayProcesses(Process p[], int n) {

    cout << "\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\n";

    float totalTAT = 0, totalWT = 0;

    for (int i = 0; i < n; i++) {
        cout << p[i].pid << "\t"
              << p[i].arrival << "\t"
              << p[i].burst << "\t"
              << p[i].priority << "\t"
              << p[i].completion << "\t"
              << p[i].turnaround << "\t"
              << p[i].waiting << endl;

        totalTAT += p[i].turnaround;
        totalWT += p[i].waiting;
    }

    cout << "\nAverage Turnaround Time: " << totalTAT / n;
    cout << "\nAverage Waiting Time: " << totalWT / n << endl;
}

int main() {

    int n;
    cout << "Enter number of processes: ";
    cin >> n;

    Process p[n];

    inputProcesses(p, n);
    priorityScheduling(p, n);
    displayProcesses(p, n);

    return 0;
}
```

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Output;

```
sions\60A96C5AC8DDCF5A4DB85F6E76326A37AFEF8938\transfers\2026-09\" ; if ($?) { g++ non_preempti
ve_priority.cpp -o non_preemptive_priority } ; if ($?) { .\non_preemptive_priority }
Enter number of processes: 4

Process 1
Arrival Time: 3
Burst Time: 4
Priority: 3

Process 2
Arrival Time: 2
Burst Time: 12
Priority: 12

Process 3
Arrival Time: 5
Burst Time: 6
Priority: 34

Process 4
Arrival Time: 7
Burst Time: 5
Priority: 57
```

PID	AT	BT	PR	CT	TAT	WT
2	2	12	12	14	12	0
1	3	4	3	18	15	11
3	5	6	34	24	19	13
4	7	5	57	29	22	17

```
Average Turnaround Time: 17
Average Waiting Time: 10.25
```

Preemtive:

```
#include <iostream>
using namespace std;

struct Process {
    int pid;
    int arrival;
    int burst;
    int remaining;
    int priority;
    int completion;
    int turnaround;
    int waiting;
};

void swap(Process &a, Process &b) {
    Process temp = a;
```

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```
    a = b;
    b = temp;
}

void sortByArrival(Process p[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (p[j].arrival > p[j + 1].arrival) {
                swap(p[j], p[j + 1]);
            }
        }
    }
}

void inputProcesses(Process p[], int n) {
    for (int i = 0; i < n; i++) {
        p[i].pid = i + 1;
        cout << "\nProcess " << p[i].pid << endl;
        cout << "Arrival Time: ";
        cin >> p[i].arrival;
        cout << "Burst Time: ";
        cin >> p[i].burst;
        cout << "Priority: ";
        cin >> p[i].priority;
        p[i].remaining = p[i].burst;
    }
}

void priorityScheduling(Process p[], int n) {

    sortByArrival(p, n);

    int currentTime = 0;
    int completed = 0;

    while (completed < n) {

        int idx = -1;
        int highestPriority = 9999;

        for (int i = 0; i < n; i++) {
            if (p[i].arrival <= currentTime && p[i].remaining > 0) {
```

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```
        if (p[i].priority < highestPriority) {
            highestPriority = p[i].priority;
            idx = i;
        }
        else if (p[i].priority == highestPriority) {
            if (p[i].remaining < p[idx].remaining) {
                idx = i;
            }
        }
    }
}

if (idx != -1) {
    p[idx].remaining--;
    currentTime++;

    if (p[idx].remaining == 0) {
        p[idx].completion = currentTime;
        p[idx].turnaround = p[idx].completion - p[idx].arrival;
        p[idx].waiting = p[idx].turnaround - p[idx].burst;
        completed++;
    }
    else {
        currentTime++;
    }
}
}
```

```
void displayProcesses(Process p[], int n) {
```

```
    cout << "\nPID\tAT\tBT\tPR\tCT\tTAT\tWT\n";
```

```
    float totalTAT = 0, totalWT = 0;
```

```
    for (int i = 0; i < n; i++) {
        cout << p[i].pid << "\t"
            << p[i].arrival << "\t"
            << p[i].burst << "\t"
            << p[i].priority << "\t"
            << p[i].completion << "\t"
            << p[i].turnaround << "\t"
            << p[i].waiting << endl;
```

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```
        totalTAT += p[i].turnaround;
        totalWT += p[i].waiting;
    }

    cout << "\nAverage Turnaround Time: " << totalTAT / n;
    cout << "\nAverage Waiting Time: " << totalWT / n << endl;
}

int main() {

    int n;
    cout << "Enter number of processes: ";
    cin >> n;

    Process p[n];

    inputProcesses(p, n);
    priorityScheduling(p, n);
    displayProcesses(p, n);

    return 0;
}
```

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```
ktop_cv1g1gvanyjgm\LocalState\sessions\60A96C5AC8DDCF5A4DB85F6E76326A37AFEF8938\transfers\202609\" ; if ($?) { g++ preemptive_priority.cpp -o preemptive_priority } ; if ($?) { .\preemptive_priority }
```

Enter number of processes: 4

Process 1

Arrival Time: 3

Burst Time: 2

Priority: 3

Process 2

Arrival Time: 4

Burst Time: 3

Priority: 2

Process 3

Arrival Time: 4

Burst Time: 5

Priority: 6

Process 4

Arrival Time: 7

Burst Time: 5

Priority: 4

PID	AT	BT	PR	CT	TAT	WT
1	3	2	3	8	5	3
2	4	3	2	7	3	0
3	4	5	6	18	14	9
4	7	5	4	13	6	1

Average Turnaround Time: 7

Average Waiting Time: 3.25